

A **paradox** is a statement or situation that seems contradictory or self-defeating but may reveal deeper insights upon closer examination.

It is a statement that, despite apparently valid reasoning from true premises, leads to a seemingly self-contradictory or a logically unacceptable conclusion.

WHAT IS BEING ASSESSED?

In critical reasoning, a paradoxical question typically assesses the test-taker's ability to identify and understand logical inconsistencies or contradictions within a given scenario or argument.

MEANING OF THE TERMS INVOLVED

Paradox

A paradox is a seemingly contradictory or self-defeating statement or situation that, upon closer examination, may reveal underlying complexities or hidden truths.

Resolution

The process of finding a way to reconcile the seemingly contradictory elements in the argument or scenario. This may involve identifying missing information or reconsidering assumptions.

TYPICAL QUESTION PATTERNS

- A passage followed by a question and one needs to identify which option helps in resolution of the paradox.

EXAMPLE QUESTION

With relaxed norms for entrepreneurs, India has come to be known as the land of opportunities, with plethora of newly formed ventures coming up and succeeding every day, especially in the post-liberalization era. However, an alarmingly high number of college graduates across all the streams still prefer working in the US or Australia after graduation.

Which of the following, if true, most helps to resolve the apparent paradox?

- Most of the college graduates prefer working abroad as their families are settled in that country.
- A college graduate doesn't prefer starting his venture unless he has got sufficient experience and exposure, and prefer working abroad for the latter.
- With the increase in the cost of college education, the graduates prefer paying back loans before starting their own venture.
- Hardly 7% of the college graduates prefer being an entrepreneur and most of them would rather go for a permanent job.

Correct option: (b)

Solution:

To resolve this paradox, the reason should be strong enough to address both the aspects of the paradox viz, entrepreneurial scenario being lucrative in India and people still preferring to go abroad. Only (b) resolves this paradox, as it shows why a graduate who wants to be an entrepreneur and start his own venture, will prefer working abroad immediately after graduation.

A **logical fallacy** is any kind of error in reasoning that renders an argument invalid. They can involve distorting or manipulating facts, drawing false conclusions, or distracting you from the issue at hand. In theory, it seems like they'd be pretty easy to spot, but this isn't always the case.

WHAT IS BEING ASSESSED?

Fallacy questions typically assess your ability to identify logical fallacies or flaws in reasoning within an argument.

MEANING OF THE TERMS INVOLVED

FLAW

A flaw in an argument refers to a defect or weakness in the reasoning that makes the conclusion less convincing or reliable. Identifying the flaw involves recognizing the error in the thought process that weakens the logical structure of the argument.

FALLACY

A fallacy is a specific type of flaw in reasoning, characterized by a common error in logic. Fallacies often lead to invalid or unsound arguments. Recognizing fallacies is essential in assessing the strength of an argument and understanding why it may be flawed.

VALID REASONING

Validity is about the logical relationship between the premises and the conclusion, ensuring that the conclusion is a logical consequence of the premises.

INVALID REASONING

Invalid reasoning often involves a flaw in the logical structure of the argument, such as a failure in the connection between the premises and the conclusion.

TYPICAL QUESTION PATTERNS

STATEMENT BASED

You will be presented with a statement or conversation between two persons and one needs to identify the flaw or fallacy in the reasoning.

EXAMPLE QUESTION:

Teacher: The mass of the sun is ten times the mass of earth.



Student: How the mass of the Sun was measured when no one reached there?

Which of the following statements best describes the flaw in the reasoning of the student?

- (a) The student's question is rooted in his imagination.
- (b) The student does not believe the statement of his teacher.
- (c) The student is expecting more explanation from the teacher.
- (d) The claims of the teacher are not validated by the coursebook.

Correct option: (a)

Solution:

Response A explains the flaw. The naiveness of this student can be explained in his imagination. He probably thinks that the mass of something can be measured through weighing machines only. His flawed thinking can be associated with his inexperience and limited knowledge.

Response B is illogical.

Response C might be correct to understand the situation but it fails to explain the flaw in the reasoning of the student.

Response D is unexpected in the behavior of a student, he is not referring to any coursebook. Students are full of imagination and they can ask any question from the teacher.

Hence, A is the answer.

• PASSAGE BASED

A passage or a group of sentences is given and one needs to identify the flaw in the reasoning or arguments presented.

EXAMPLE QUESTION:

People feel stressed as they hear about interviews. Most of them focus only on the questions to be answered in the interview. But the interview is the test of not only knowledge but behavior and honesty too. It does not matter where you went to school, the number of degrees you may hold, the experience you have, or whom you know.

The argument given above is flawed in –

- (a) Ignoring the importance of competencies.
- (b) Overlooking the importance of experience.
- (c) Not discussing the role of alma mater and peer group.
- (d) Not giving importance to the right attitude.

Correct option: (a)

Solution:

Response A is correct. Every job needs some skills and education. An interview is conducted to find a suitable candidate with these skills and qualifications. Without these, he will not be able to perform the job efficiently. So, the value of competency cannot be overlooked.

Response B is a sub-set of option A.

Response C is not logical in the general context. The role of the alma mater can be understood but the role of a peer group is very specific to some high-profile management jobs only.

Response D is incorrect as it is a supporting statement and not a weakening statement for the argument presented in the passage.

Hence, A is the correct answer.



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SKILL DRILL

Directions (Q.1-Q.25): Read the passage given below and answer the question that follows

1. Time and again it has been shown that students who attend colleges with low faculty/student ratios get the most well-rounded education. As a result, when my children are ready to attend college, I'll be sure they attend a school with a very small student population. **Which of the following, if true, identifies the greatest flaw in the reasoning above?**

- (a) A low faculty/student ratio is the effect of a well-rounded education, not its source.
- (b) Intelligence should be considered the result of childhood environment, not advanced education.
- (c) A very small student population does not by itself, ensure a low faculty/student ratio.
- (d) Parental desires and preferences rarely determine a child's choice of a college or university.

2. The usefulness of lie detectors cannot be overestimated. Although there is no employee screening procedure that is 100 percent accurate, the lie detector is a valuable tool for employers and employees alike. The lie detector's usefulness is amply demonstrated in a recent survey conducted by a prestigious university. In the survey, those employees of a large company who were applying for a newly created position within the company were asked if they had ever worked on Project X. More than one-third of the applicants studied lied and said they had worked on the project—a project that never existed.

Which one of the following best identifies a flaw in the author's argument about the usefulness of lie detectors?

- (a) The argument depends on the assumption that whatever is good for the employer is good for the employee.
- (b) Since lie detectors are known to be less than 100 percent accurate, the test will tend to help only those with something to hide.
- (c) By referring to a prestigious university, the author is appealing to authority rather than to evidence.
- (d) The study shows only that certain individual will lie, not that the lie detector can detect them.

3. **Alba:** I don't intend to vote for Senator Frank in the next election. She is not a strong supporter of the war against crime.

Tam: But Senator Frank sponsored the latest anticrime law passed by the Senate.

Alba: If Senator Frank sponsored it, it can't be a very strong anticrime law.

Which of the following identifies the most serious logical flaw in Alba's reasoning?

- (a) The facts she presents do not support her conclusion that Senator Frank is soft on crime.
- (b) She assumes without proof that crime is the most important issue in the upcoming election.
- (c) She argues in a circle, using an unsupported assertion to dismiss conflicting evidence.
- (d) She attacks Senator Frank on personal grounds rather than on his merit as a political leader.

4. This year the New Hampshire Division of Company X set a new record for annual sales by that division. This record is especially surprising since the New Hampshire Division has the smallest potential market and the lowest sales of any of Company X's divisions.

Which of the following identifies a flaw in the logical coherence of the statement above?

- (a) If overall sales for Company X were sharply reduced, the New Hampshire Division's new sales record is irrelevant to the company's prosperity.
- (b) Since the division is competing against its own record, the comparison of its sales record with that of other divisions is irrelevant.
- (c) If this is the first year that the New Hampshire Division has been last in sales among Company X's divisions, the new record is not surprising at all.
- (d) If overall sales for Company X were greater than usual, it is not surprising that the New Hampshire Division was last in sales.

5. The average normal infant born in the United States weighs between twelve and fourteen pounds at the age of three months. Therefore, if a three-month-old child weighs only ten pounds, its weight gain has been below the United States average.

Which of the following indicates a flaw in the reasoning above?

- (a) Weight is only one measure of normal infant development.
- (b) Some three-month-old children weigh as much as seventeen pounds.
- (c) It is possible for a normal child to weigh ten pounds at birth.
- (d) The phrase "below average" does not necessarily mean insufficient.

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6. A discount retailer of basic household necessities employs thousands of people and pays most of them at the minimum wage rate. Yet following a federally mandated increase of the minimum wage rate that increased the retailer's operating costs considerably, the retailer's profits increased markedly.

Which of the following, if true, most helps to resolve the apparent paradox?

- (a) Over half of the retailer's operating costs consist of payroll expenditures; yet only a small percentage of those expenditures go to pay management salaries.
 - (b) The retailer's customer base is made up primarily of people who earn, or who depend on the earnings of others who earn, the minimum wage.
 - (c) The retailer's operating costs, other than wages, increased substantially after the increase in the minimum wage rate went into effect.
 - (d) When the increase in the minimum wage rate went into effect, the retailer also raised the wage rate for employees who had been earning just above minimum wage.
7. The average level of fat in the blood of people suffering from acute cases of disease W is lower than the average level for the population as a whole. Nevertheless, most doctors believe that reducing blood-fat levels is an effective way of preventing acute W.

Which one of the following, if true, does most to justify this apparently paradoxical belief?

- (a) The blood level of fat for patients who have been cured of W is on average the same as that for the population at large.
 - (b) Several of the symptoms characteristic of acute W have been produced in laboratory animals fed large doses of a synthetic fat substitute, though acute W itself has not been produced in this way.
 - (c) The progression from latent to acute W can occur only when the agent that causes acute W absorbs large quantities of fat from the patient's blood.
 - (d) The levels of fat in the blood of patients who have disease W respond abnormally slowly to changes in dietary intake of fat.
8. Some companies in fields where skilled employees are hard to find make signing an "agreement not to compete" a condition of employment. In such an agreement the employee promises not to go work for a competing firm for a set period after leaving his or her current employer. Courts are increasingly ruling that these agreements are not binding. Yet paradoxically, for people who signed such agreements when working for competing firms, many firms are unwilling to

consider hiring them during the period covered by the agreement.

Which one of the following, if true, most helps to resolve the paradox?

- (a) Many companies will not risk having to become involved in lawsuits, even suits that they expect to have a favorable outcome.
 - (b) In some industries, for example the broadcast media, companies' main source of new employees tends to be people who are already employed by competing firms.
 - (c) Most companies that require their employees to sign agreements not to compete are aware that these documents are not legally binding.
 - (d) Many people who have signed agreements not to compete are unwilling to renege on a promise by going to work for a competing firm.
9. Police statistics have shown that automobile anti-theft devices reduce the risk of car theft, but a statistical study of automobile theft by the automobile insurance industry claims that cars equipped with anti-theft devices are, paradoxically, more likely to be stolen than cars that are not so equipped.

Which one of the following, if true, does the most to resolve the apparent paradox?

- (a) Owners of stolen cars almost invariably report the theft immediately to the police but tend to delay notifying their insurance company, in the hope that the vehicle will be recovered.
 - (b) Most cars that are stolen are not equipped with anti-theft devices, and most cars that are equipped with anti-theft devices are not stolen.
 - (c) The most common automobile anti-theft devices are audible alarms, which typically produce ten false alarms for every actual attempted theft.
 - (d) Automobile owners who have particularly theft-prone cars and live in areas of greatest incidence of car theft are those who are most likely to have anti-theft devices installed.
10. Since the introduction of the Impanian National Health Scheme, Impanians (or their private insurance companies) have had to pay only for the more unusual and sophisticated medical procedures. When the scheme was introduced, it was hoped that private insurance to pay for these procedures would be available at modest cost, since the insurers would no longer be paying for the bulk of health care costs, as they had done previously. Paradoxically, however, the cost of private health insurance did not decrease but has instead increased dramatically in the years since the scheme's introduction.



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Which one of the following, if true, does most to explain the apparently paradoxical outcome?

- (a) The National Health scheme has greatly reduced the number of medical claims handled annually by Impania's private insurers, enabling these firms to reduce overhead costs substantially.
- (b) Before the National Health scheme was introduced, more than 80 percent of all Impanian medical costs were associated with procedures that are now covered by the scheme.
- (c) Impanians who previously were unable to afford regular medical treatment now use the National Health scheme, but the number of Impanians with private health insurance has not increased.
- (d) Impanians now buy private medical insurance only at times when they expect that they will need care of kinds not available in the National Health scheme.

11. Since 1945 pesticide use in the United States has increased tenfold despite an overall stability in number of acres planted. During the same period, crop loss from insects has approximately doubled, from about seven to thirteen percent.

Which one of the following, if true, contributes most to explaining the paradoxical findings above?

- (a) Extension agents employed by state governments to advise farmers have recently advocated using smaller amounts of pesticide, though in past years they promoted heavy pesticide use.
- (b) While pesticide-resistant strains of insects were developing, crop rotation, which for insects disrupts a stable food supply, was gradually abandoned because farmers' eligibility to receive government crop subsidies depended on continuing to plant the same crop.
- (c) Since 1970 the pesticides most lethal to people have generally been replaced by less-lethal chemicals that are equally effective against insects and have a less-damaging effect on the fish in streams fed by water that runs off from treated agricultural fields removed from foodstuffs through washing, others are taken up into the edible portion of plants, and consumers have begun to boycott foods containing pesticides that cannot be washed off.
- (d) Because farmers' decisions about how much land to plant are governed by their expectations about crop prices at harvest time, the amount of pesticide they apply also depends in part on expected crop prices.

12. Good body, Inc., is in the process of finding tenants for its newly completed Parrot Quay commercial development, which will make available hundreds of thousands of square feet of new office space on what

was formerly derelict property outside the financial center of the city. Surprisingly enough, the coming recession, though it will hurt most of the city's businesses, should help Good body to find tenants.

Which one of the following, if true, does most to help resolve the apparent paradox?

- (a) Businesses forced to economize by the recession will want to take advantage of the lower rents available outside the financial center.
- (b) Public transportation links the financial center with the area around Parrot Quay.
- (c) The area in which the Parrot Quay development is located became derelict after the heavy industry that used to be there closed down in a previous recession.
- (d) Many of Good body's other properties are in financial center and will become vacant if the recession is severe enough to force Good body's tenants out of business.

13. In a critical discussion on economic reforms, Ms. Kapoor, a respected economist, subtly stated, "Mr. Singh's suggestion to increase taxes on luxury goods to boost the economy is an interesting perspective. However, considering Mr. Singh's well-known penchant for luxury goods, one might wonder if his personal bias could be clouding his judgment. Therefore, we should carefully evaluate the potential implications of his suggestion before making any decisions."

Which of the following best identifies a flaw in Ms. Kapoor's reasoning?

- (a) Ms. Kapoor fails to consider that an economist's personal preferences do not necessarily invalidate their professional judgment on economic matters.
- (b) Ms. Kapoor does not provide statistical data to support the claim that increasing taxes on luxury goods will boost the economy.
- (c) Ms. Kapoor overlooks the possibility that other economists might also have personal biases that could influence their professional suggestions.
- (d) Ms. Kapoor does not acknowledge that increasing taxes on luxury goods might have negative implications for the economy.

14. The great Roman Julius Caesar was hated by all Roman citizens. This animosity is shown by historians through multiple documents describing the circumstances surrounding Caesar's death. These documents reveal that in the end even his best friend, Brutus, wanted to kill him. Roman politicians and senators also conspired to bring about his death.

A flaw in the argument is that the author



- (a) assumes that whatever is true of a group is true of a sample of that group
- (b) argues for a course of action that is counterproductive
- (c) makes a conclusion based on a small and unrepresentative sample of a group
- (d) makes an appeal to the opinion of an implausible authority
15. "In a recent survey, 90% of the city's residents reported satisfaction with public transportation. The survey was conducted on a weekday morning, between 9 am and 12 pm, at the city's main bus terminal. The results clearly indicate that the city's public transportation system is meeting the needs of its residents."
- Which of the following best identifies the flaw in this reasoning?**
- (a) The conclusion is based on a survey conducted at a specific location (the city's main bus terminal), which may not represent the views of all residents in the city.
- (b) The argument fails to consider that satisfaction with public transportation might vary at different times of the day and on weekends.
- (c) The reasoning overlooks the possibility that people who are dissatisfied with public transportation may not use it and thus were not present to be surveyed.
- (d) The argument assumes that satisfaction with public transportation is the only indicator of the system meeting the residents' needs.
16. Given the increasing traffic congestion, the city must either build more roads or face economic decline. The city's future prosperity hinges on this crucial decision.
- Which of the following best identifies the flaw in this reasoning?**
- (a) The argument assumes a direct causal relationship between building more roads and solving traffic congestion without considering other factors that contribute to traffic problems.
- (b) The reasoning overlooks alternative solutions to traffic congestion, such as improving public transportation or implementing traffic management systems.
- (c) The argument fails to consider the potential negative environmental and social impacts of building more roads on the city and its residents.
- (d) The reasoning assumes that economic decline is the only possible consequence of not building more roads, ignoring other potential outcomes of traffic congestion.
17. In a televised debate, a politician argues that since a country with a very high GDP has an excellent public healthcare system, improving the economy of a country with a lower GDP will automatically lead to a better healthcare system there. This politician points out that economic growth is the primary driver of healthcare quality, based on the example of the high-GDP country.
- How does the politician's reasoning potentially oversimplify the relationship between GDP and healthcare quality?**
- (a) By assuming that economic prosperity is the only factor influencing healthcare standards.
- (b) By overlooking different healthcare policies and infrastructures in other countries.
- (c) By failing to consider that healthcare advancements might occur independently of global growth.
- (d) By suggesting a singular approach to healthcare improvement, ignoring other viable methods.
18. Since the launch of our new security measures last month, there have been no reports of unauthorized access in our facilities. We can safely say that our new security measures are effectively deterring any potential intruders.
- Which of the following best identifies the flaw in this reasoning?**
- (a) The argument assumes that the effectiveness of the security measures can be accurately assessed within a short period of time.
- (b) The reasoning relies on the absence of evidence (no reports of unauthorized access) as definitive proof of the security measures' effectiveness.
- (c) The argument fails to consider other factors that might have contributed to the lack of unauthorized access reports, such as reduced operational activities.
- (d) The reasoning does not take into account the possibility that unauthorized access could occur in all ways detectable by the new security measures.
19. "We can say for certain that our company is the industry leader. The sole reason for that is we are the most innovative. Now how do we know this? How do we do say for sure that we innovate like no other? Well, that is because, as the industry leader, we set the trends that others follow."
- Which of the following best identifies the flaw in this reasoning?**
- (a) The claim that the company is the industry leader is both a premise and a conclusion, supported by the assertion of innovation, which is in turn supported by being the industry leader.
- (b) The reasoning fails to provide empirical evidence or external validation to support the claim that the company is the most innovative in the industry.



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- (c) The argument overlooks the possibility that other companies in the industry might also be setting trends and contributing to innovation.
- (d) The reasoning does not consider that being the industry leader does not necessarily mean that the company is the most innovative.

20. A software development company, XYZ Corp, has been striving to enhance its efficiency and employee satisfaction. In the past year, they have observed a notable improvement in overall employee productivity, measured by project completion rates and client satisfaction scores. Interestingly, around the same time they started noticing this improvement, XYZ Corp had implemented a policy change - they switched from a traditional five-day workweek to a more flexible four-day workweek model. The company's management team is now convinced that this significant increase in productivity is directly and solely due to the introduction of the four-day workweek. They argue that the shorter workweek has led to more focused and efficient work from their employees, thereby boosting productivity.

Which of the following best identifies the flaw in this reasoning?

- (a) The argument commits a fallacy by assuming that the implementation of the four-day workweek is the direct cause of increased productivity, without considering other potential contributing factors.
- (b) The reasoning fails to account for the possibility that improved project completion rates and client satisfaction scores could be due to factors related to workweek length, such as market trends or changes in client demands.
- (c) The argument overlooks the potential long-term effects of a four-day workweek on employee productivity, focusing only on the immediate outcomes observed in the past year.
- (d) The management presumes that employee satisfaction is directly proportional to productivity, ignoring other aspects that contribute to an employee's overall job satisfaction.

21. There are only two kinds of fruits that can be grown on a large scale in Nashvin County-bananas and oranges. the unique climate of Nashville County is of particular benefit to the orange trees and as a result, the total number of oranges that can be grown in one year in Nashville County is three times the number of bananas that can be grown in a year. However, most orchard owners still focus on growing more banana trees than orange trees.

Which of the following best explains the paradox highlighted above?

- (a) While the demand for oranges in the domestic market is low, the export demand for oranges is huge.
- (b) the selling price of a dozen bananas is at least 6 times that of a dozen oranges.
- (c) the amount of fertilizer required by banana trees and orange trees is more or less the same.
- (d) Most orchard owners in other neighboring counties also focus more on growing bananas than on growing oranges.

22. NTPC has brought down 16 million watts of power generation for want of demand. On the other hand several north eastern states are reeling in darkness due to severe shortage in power supply.

Which of the following, if true, explains the paradox in the above situation?

- (a) The north eastern states have not cleared their power bills payable to NTPC for the past two years.
- (b) Supply cables in north eastern states have been damaged to a large extent during heavy monsoon rains.
- (c) NTPC has also brought down three millions of power generation last year for the same reason, while north eastern states received sufficient power supply during the same period.
- (d) NTPC supplies power to southern states only, while north eastern states receive supply from Northern Grid.

23. Most broken bones need to be treated and set by a doctor in order to heal properly. Otherwise, broken bones can ossify in the wrong places, and injured people will have limited functionality in the once-broken bones for the rest of their lives. However, some doctors recommend against any treatment for broken bones in certain appendages.

Which of the following, if true, explains this paradox?

- (a) Some doctors believe that bones in certain appendages heal better when left untreated.
- (b) Getting medical treatment for a broken bone often disturbs the healing process.
- (c) Sometimes surgery can cause bones to heal less well than they would have without the surgery.
- (d) Many people would rather not have surgery if at all possible.



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24. Scientists estimate that the Colombian rainforest holds about 100 million hectares of forestland. According to a survey based solely on models of the current rate of rainforest depletion, in 25 years' time all of this forestland will be gone. However, despite these models, Colombian scientists claim that there will be no more rainforests left in Colombia in as few as 15 years.

Which of the following, if true, helps to best reconcile the scientists' conclusion with the survey's projection?

- (a) Colombian scientists have a vested interest in increasing global awareness of the deforestation occurring in the Colombian rainforest.
 - (b) Colombian scientists take into account the fact that future technologies will probably speed up the rate of depletion.
 - (c) Farmers and loggers contribute to the current rate of deforestation, but farmers will not contribute to future rates.
 - (d) Rainforest regrowth will reduce the overall depletion rate if conservation groups actively plant new seedlings.
25. Most oil is transported around the world by sea in large oil barges. Along with this travel comes the small risk of a shipwreck that would spill the oil into the ocean, destroying the habitat of many sea creatures. The waters near Spain have had two major oil spills in the last year and a half. This summer, a third oil barge called to Spanish harbors requesting port since it was badly spilling oil. The captain wanted to stop this spill immediately, but even though the Spanish government highly values preventing oil spills, it did not allow the barge to dock and forced the captain to turn further out to sea.

Which of the following, if true, most explains the actions of the Spanish government?

- (a) It costs more to clean up a spill that is far from the coast than a spill that is nearer to the coast.
- (b) Most staunch environmentalists would have immediately given the barge permission to dock in a Spanish port.
- (c) A leaking barge journeying to a Spanish port would destroy more sea life near Spain than one turning away to deeper waters.
- (d) The barge was not spilling badly when it made the call, but the spill would eventually worsen if left untreated.



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